

1. Introduction

The objective of this project was to analyze borrower and loan characteristics and build predictive models for loan default status.

I first performed exploratory data analysis to understand the data and identify potential risk factors, then developed and compared Logistic Regression, Random Forest, and XGBoost models to evaluate predictive performance.

2. Dataset Description

The dataset contains 58,645 loan records and 13 variables. The target variable is *loan_status*, where 1 means a defaulted loan and 0 means a non-default loan.

Following the project requirements, I removed the *id* column and treated all remaining variables as predictors. The dataset includes demographic information (age), financial characteristics, credit history, and loan-related variables. These features provide useful information about people's repayment ability and credit risk.

3. Exploratory Data Analysis (EDA)

(1) Data Quality Check

I first performed several basic data quality checks and found that the dataset was relatively clean, with no missing values or duplicated records.

I also identified several suspicious extreme values. Specifically, *person_age* and *person_emp_length* contained unrealistic values of 123 years. Since these values were clearly unreasonable and likely reflected data errors, I removed those observations from the dataset to avoid distortion in the analysis and model training.

(2) Target Variable Distribution

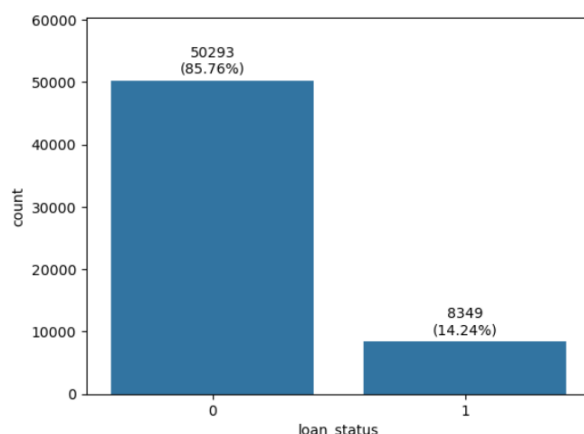


Figure 1 Loan Status Distribution

The dataset was imbalanced, with approximately 14.24% default loans and 85.76% non-default loans. As a result, Accuracy alone could be misleading. Therefore, I

focused primarily on Precision, Recall, F1-score, and ROC-AUC when evaluating model performance.

(3) Independent Variables Analysis

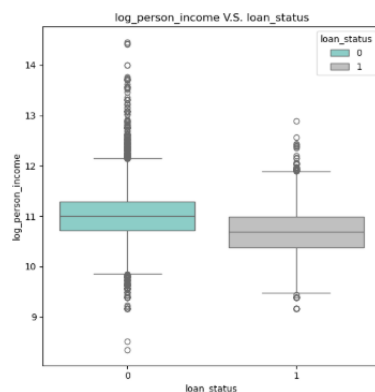


Figure 2 Log Personal Income V.S. Loan Status

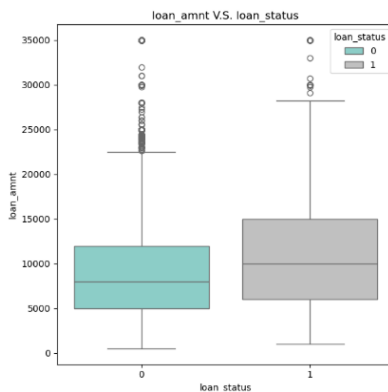


Figure 3 Loan Amount V.S. Loan Status

Borrowers with lower incomes and larger loan amounts generally had higher default rates, although loan burden was captured more clearly by *loan_percent_income* than by income and loan amount separately.

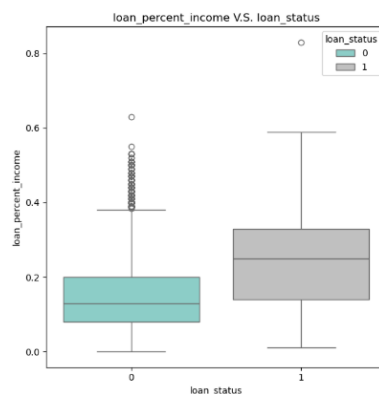


Figure 4 Loan burden V.S. Loan Status

Loan burden was one of the most critical risk indicators. People with higher *loan_percent_income* values were more likely to default, implying that repayment pressure plays an important role in credit risk.

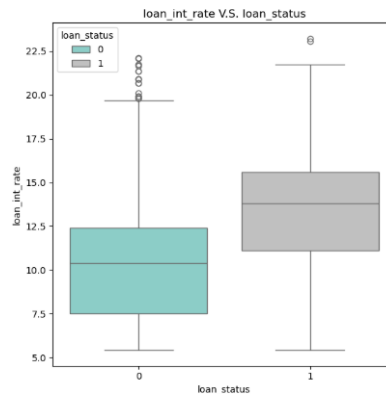


Figure 5 Loan Interest Rate V.S. Loan Status

Table 1 Loan Interest Rate Group V.S. Default Rate

loan_int_group	sample_size	default_rate
<8%	17596	4.540805
8%-12%	21766	9.478085
12%-16%	17106	24.640477
>16%	2174	58.509660

Higher interest rates (*loan_int_rate*) were strongly associated with higher default rates. However, this relationship likely reflects risk-based pricing, as lenders typically assign higher interest rates to people with lower credit quality (*loan_grade*) or previous default histories (*cb_person_default_on_file*).

To better illustrate how default risk changes, loan interest rates were grouped into four interpretable ranges. Default rates increased substantially across these groups, rising from 4.5% among rates below 8% to nearly 59% among rates above 16%.

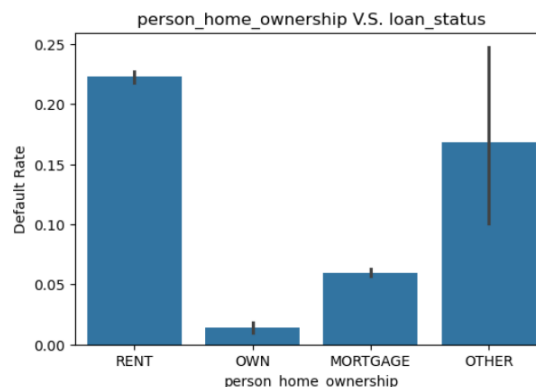


Figure 6 Home Ownership V.S. Loan Status

Table 2 Home Ownership V.S. Default Rate

person_home_ownership	sample_size	default_rate
OWN	3138	1.370300
MORTGAGE	24821	5.970751
OTHER	89	16.853933
RENT	30594	22.255998

Home ownership status appeared to be strongly associated with default risk. Renters had a higher probability of default than homeowners or people who held mortgages.

Although the *OTHER* category showed a relatively high default rate, the sample size was very small and should be interpreted cautiously.

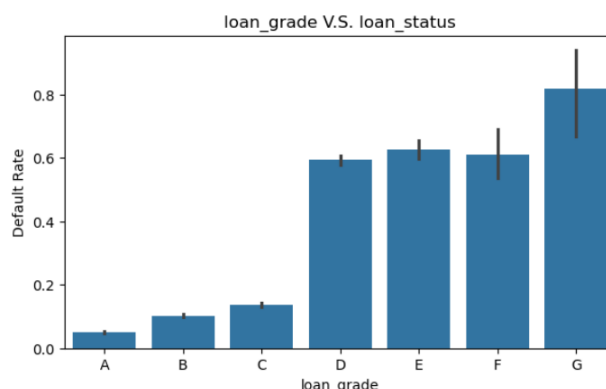


Figure 7 Loan Grade V.S. Loan Status

Table 3 Loan Grade V.S. Default Rate

loan_grade	sample_size	default_rate
A	20984	4.918033
B	20398	10.231395
C	11036	13.537514
D	5033	59.348301
E	1009	62.537166
F	149	61.073826
G	33	81.818182

Likelihood of defaulting generally increased from grade *A* to *G*. This pattern suggested that *loan_grade* was an effective measure of credit risk. However, the results for grades *E*, *F*, and *G* should be interpreted carefully because those groups contain relatively few observations.

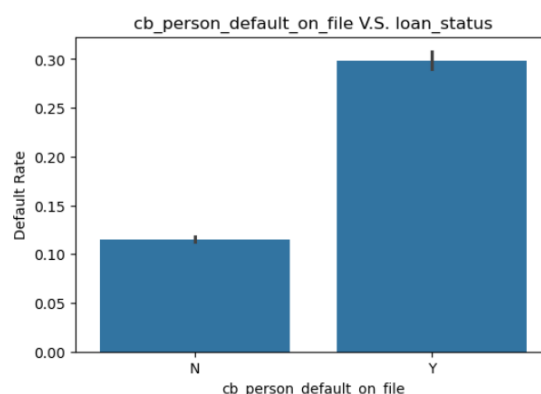


Figure 8 Previous Default Record V.S. Loan Status

Table 4 Previous Default Record V.S. Default Rate

cb_person_default_on_file	sample_size	default_rate
N	49941	11.511584
Y	8701	29.881623

People with previous default records (*cb_person_default_on_file* = 1) showed a greater tendency to default than people without records. This finding suggested that

historical repayment behavior contains valuable information about future default risk. However, much of this information appeared to be incorporated into loan grade.

(4) Key Findings

The main findings from the exploratory data analysis are summarized below:

- The dataset contained no missing values or duplicated records.
- The target variable was imbalanced, with approximately 14.24% default loans and 85.76% non-default loans.
- Unrealistic observations for age and employment length were removed.
- Several variables were strongly associated with default risk, including loan burden (*loan_percent_income*), interest rates (*loan_int_rate*), loan grade (*loan_grade*), and previous default history (*cb_person_default_on_file*). In addition, many of them appeared to capture overlapping information, which I examined further in the modeling stage.
- Renters showed higher default rates than homeowners or people with mortgages.

4. Modeling Methodology

(1) Data Preprocessing and Feature Engineering

Based on the findings from the EDA section, I performed several data preprocessing and feature engineering steps before model training.

First, I applied a log transformation to *person_income*. The distribution of *person_income* was highly right-skewed with several extremely high-income observations. Because these values may represent genuine high-income people rather than data entry errors, I used a log transformation to reduce skewness and improve model stability without removing potentially valid observations.

Second, I created a centered squared age term (*person_age_sq*), allowing the model to capture potential nonlinear effects of age on default risk.

Third, categorical variables were encoded before model training. Variables such as *person_home_ownership*, *loan_intent*, and *loan_grade* were converted into dummy variables using one-hot encoding, while *cb_person_default_on_file* was converted into a binary indicator (0 = No, 1 = Yes).

In addition, numerical variables were standardized to bring all features to the same scale when training Logistic Regression models. This improves optimization stability and helps the model converge more efficiently during training.

After preprocessing, the resulting features were used for multicollinearity analysis and subsequent model development.

(2) Multicollinearity

Before model training, I evaluated multicollinearity among predictors using the Variance Inflation Factor (VIF).

Table 5 Variance Inflation Factor

Feature	VIF
loan_int_rate	10.8257
C	8.5879
D	7.4123
person_age	5.8750
MORTGAGE	5.3802
RENT	5.2734
B	4.9329
cb_person_cred_hist_length	4.3575
E	2.9202
person_age_sq	2.0600
cb_person_default_on_file	1.7157
MEDICAL	1.5683
PERSONAL	1.5319
VENTURE	1.5249
DEBTCONSOLIDATION	1.4931
HOMEIMPROVEMENT	1.4125
F	1.3946
log_person_income	1.3354
loan_percent_income	1.1626
person_emp_length	1.1390
G	1.1155
OTHER	1.0280

Most predictors exhibited acceptable VIF values. However, *loan_int_rate* showed substantial multicollinearity, with a VIF value of 10.83. This result was expected because people with lower credit quality were generally assigned both lower loan grades and higher interest rates. As a result, loan grade and interest rate contained highly overlapping information.

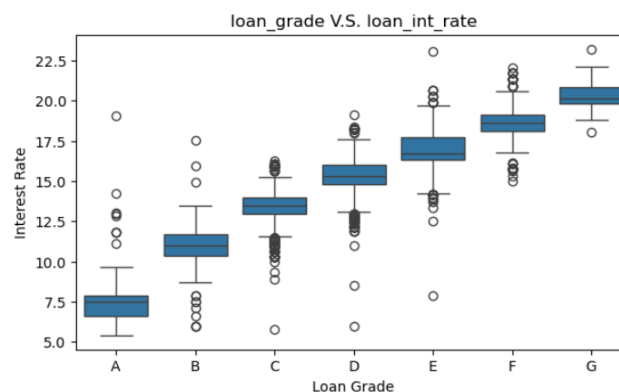


Figure 9 Loan Grade V.S. Loan Interest Rate

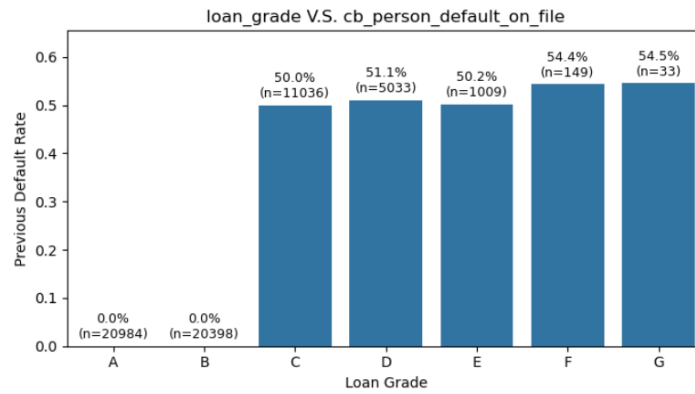


Figure 10 Loan Grade V.S. Previous Default Rate

People with worse loan grades were generally assigned higher interest rates, explaining the high VIF value observed for *loan_int_rate*. Besides, previous default records were concentrated primarily in lower loan grades. This suggested that loan grade already reflected the information related to historical default behavior.

In addition to VIF analysis, coefficient significance tests from Logistic Regression models provided further support for feature selection. When *loan_grade* was included in the model, *cb_person_default_on_file* was no longer statistically significant, showing that loan grade already captured most of the predictive information contained in previous default records. Similarly, *cb_person_cred_hist_length* was not statistically significant after controlling for *person_age*, indicating conceptual overlap between these variables.

Based on these findings, several predictors were removed to reduce redundancy and improve model interpretability.

- Credit record length (*cb_person_cred_hist_length*) was removed because it is naturally correlated with borrower age.
- Previous default record (*cb_person_default_on_file*) was removed because its predictive information was largely reflected in loan grade.
- Interest rate (*loan_int_rate*) was removed because it is highly correlated with loan grade and provides overlapping information.

I retained *loan_grade* instead of *loan_int_rate* because it provides a more direct summary of credit quality and is easier to interpret. Interest rate may also be affected by external factors such as market conditions and lender's pricing policies.

The final model utilized 19 engineered features, including *log_person_income*, *person_age* and *person_age_sq*, *person_emp_length*, *person_home_ownership* dummies, *loan_intent* dummies, *loan_grade* dummies, and *loan_percent_income*.

(3) Model Selection and Evaluation Metrics

A. Logistic Regression

I used Logistic Regression as the baseline model because it is simple, interpretable, and commonly used for binary classification problems. The models included demographic characteristics, home ownership status, loan purpose, loan burden indicators, and either loan grade or interest rate. It provided a simple benchmark for comparison against more advanced machine learning models.

Because *loan_grade* and *loan_int_rate* contained overlapping information about default risk, I constructed two separate Logistic Regression models. The first model used loan grade, while the second replaced loan grade with interest rate. This design allowed a clearer comparison of their predictive value while reducing multicollinearity concerns. The EDA findings and coefficient significance tests from the Logistic Regression models were jointly used to determine the final model. Variables that became statistically insignificant after controlling for related predictors were excluded, while variables with stronger predictive value and better interpretability were retained.

Because the dataset was imbalanced, *class_weight = "balanced"* was applied during model training to assign greater importance to the minority default class.

B. Tree-based Models

To further improve predictive performance, I also implemented two tree-based machine learning models, Random Forest and XGBoost.

(A) Random Forest

Random Forest was used to capture nonlinear relationships and complex interactions among variables while being relatively robust to overfitting.

Before model training, hyperparameters were optimized using *RandomizedSearchCV* with five-fold cross-validation to improve model performance and reduce overfitting. The final model used 100 trees with a maximum depth of 15, providing a balance between predictive performance and overfitting control.

Similar to Logistic Regression, *class_weight = "balanced"* was applied during model training to address class imbalance.

(B) XGBoost

XGBoost was implemented because it can effectively capture nonlinear relationships and complex feature interactions. It is often one of the better-performing models for structured tabular data and also provides feature

importance scores that help interpret model predictions.

As with the Random Forest model, hyperparameters were optimized using RandomizedSearchCV with five-fold cross-validation before training. The final model used 300 boosting rounds with a learning rate of 0.1, a maximum depth of 5. Both the row subsampling ratio and column subsampling ratio were set to 0.9 to improve generalization and reduce overfitting.

Because the dataset was imbalanced, *scale_pos_weight* was used to place greater emphasis on the minority default class during training.

C. Evaluation Metrics

The performance of all models was evaluated and compared using Accuracy, Precision, Recall, F1-score, and ROC-AUC. ROC-AUC was treated as the primary evaluation metric because it evaluates the model's ability to distinguish between default and non-default cases across all classification thresholds and is less affected by class imbalance than Accuracy.

5. Results and Model Comparison

(1) Logistic Regression

Table 6 Logistic Regression Model Performance

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression (loan_grade)	0.8383	0.4624	0.8311	0.5942	0.9043
Logistic Regression (loan_int_rate)	0.7984	0.3996	0.8281	0.5391	0.8869
Logistic Regression	0.8391	0.4637	0.8311	0.5953	0.9043

The model using *loan_grade* achieved a higher ROC-AUC score (0.9043) than the model using *loan_int_rate* (0.8869), suggesting that loan grade provided a more comprehensive representation of borrower credit quality.

Based on these results, *loan_grade* was retained in the final modeling framework while *loan_int_rate* was excluded. In addition, *cb_person_cred_hist_length* and *cb_person_default_on_file* were removed because their predictive information was largely captured by age-related variables and loan grade.

The final Logistic Regression model achieved a ROC-AUC score of **0.9043**, demonstrating strong predictive performance despite its relatively simple structure.

(2) Random Forest

Table 7 Random Forest Model Performance

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Random Forest	0.9453	0.8512	0.7467	0.7955	0.9417

The Random Forest model achieved a ROC-AUC score of **0.9417**, representing a notable improvement over the baseline Logistic Regression model. This improvement suggests that the relationships between predictors and loan status may not be fully captured by a simple linear model.

(3) XGBoost

Table 8 XGBoost Model Performance

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
XGBoost	0.9318	0.7238	0.8425	0.7786	0.9600

The XGBoost model achieved the highest predictive performance, with a ROC-AUC score of **0.9600**, outperforming both Logistic Regression and Random Forest.

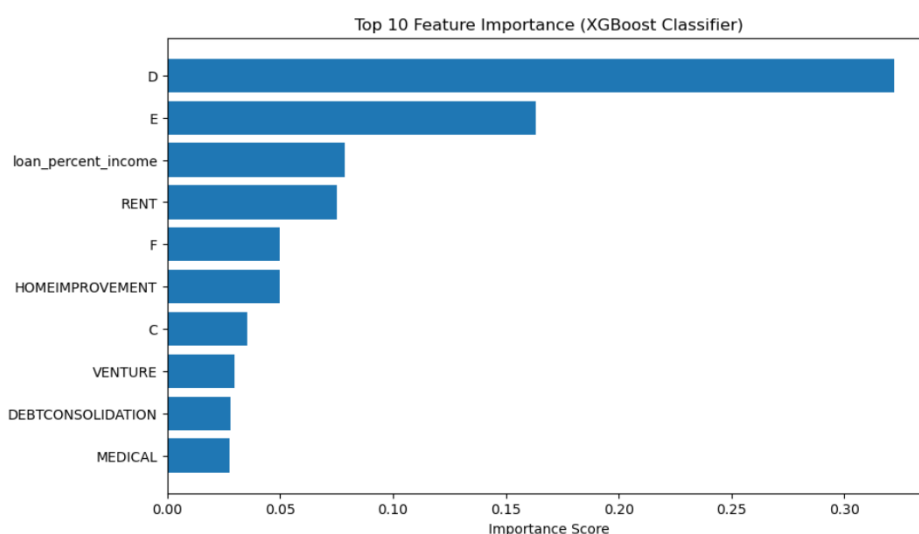


Figure 11 Top 10 Features for Loan Status Prediction

Based on the XGBoost feature importance analysis, the most important predictors of default risk were:

- **Loan Grade:** Loan grades D, E, and F were among the strongest predictors of default risk. Although Grade G showed lower importance, the result should be interpreted cautiously because it contained very few observations.
- **Financial Burden:** The *loan_percent_income* variable was one of the most important predictors. People with higher loan-to-income ratios were more likely to default.
- **Housing Status:** People who rented their homes (*RENT*) showed higher default risk than homeowners.

These findings were consistent with the EDA results, especially the strong effects of loan burden, loan grade, and housing status.

(4) Model Comparison

Table 9 Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	0.8391	0.4637	0.8311	0.5953	0.9043
Random Forest	0.9453	0.8512	0.7467	0.7955	0.9417
XGBoost	0.9318	0.7238	0.8425	0.7786	0.9600

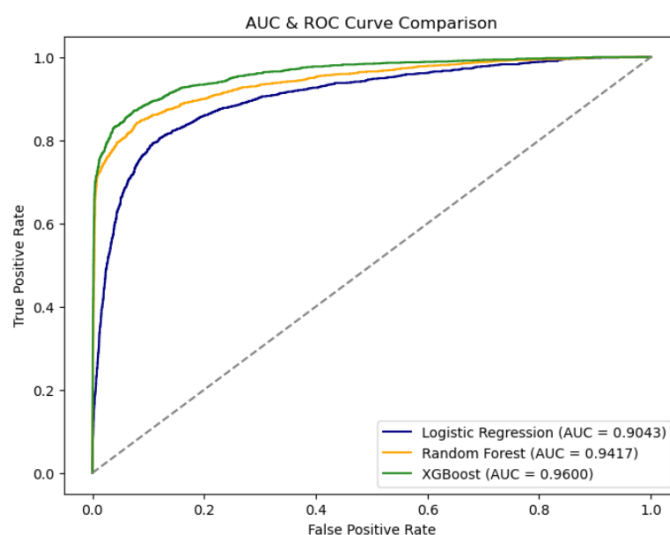


Figure 12 ROC Curve Comparison

Among all models, XGBoost achieved the highest ROC-AUC score (0.9600) while maintaining strong Recall and overall classification performance, indicating the strongest ability to distinguish between default and non-default borrowers across different classification thresholds.

Random Forest performed exceptionally well and achieved the highest Accuracy, Precision, and F1-score. The high Precision indicated that Random Forest generated fewer false positive predictions than the other models while maintaining strong overall performance.

Although Logistic Regression produced lower predictive performance than the tree-based models, it remained a valuable baseline because of its simplicity and interpretability.

Overall, the results showed that tree-based machine learning models outperformed Logistic Regression for loan default prediction. The performance gains from Random Forest and XGBoost suggested that more flexible modeling approaches may better capture the underlying relationships in the data.

6. Conclusion

This project analyzed borrower and loan characteristics and developed predictive models for loan default risk. The EDA results revealed that people with higher loan-to-income ratios, poorer loan grades, and rental housing status were more likely to default.

Three supervised learning models were developed and evaluated, including Logistic Regression, Random Forest, and XGBoost. While Logistic Regression provided a useful and interpretable baseline, both tree-based models achieved substantially better predictive performance. Among all models, XGBoost achieved the highest ROC-AUC score (0.9600). Feature importance analysis further confirmed that credit quality (*loan_grade*), financial burden (*loan_percent_income*), and housing status (*RENT*) were among the most influential predictors of loan status. These findings may help financial institutions improve credit risk assessment and identify potentially high-risk borrowers more effectively.

Future work may include additional feature engineering and neural network models to further improve predictive performance.